

1 Proofs & Counterexamples, § 4, 5, 9

1. Prove or give a counterexample to each of the following “existential statements”.

(a) There exists a prime number p with $90 < p < 105$.

(b) There exists a real number r with $r^2 + 2r + 2 \leq 0$.

(c) There exists a sequence of 100 consecutive composite numbers.

2. Prove or give a counterexample to each of the following “universal statements”.

(a) For every real number r , $r^2 + 2r + 1 \geq 0$.

(b) For all positive integers n , the number $\frac{n(n+1)}{2}$ is an integer.

(c) Every prime number is odd.

Prove or give a counterexample to each of the following statements.

3. Let a and b be integers. If $a \mid b$, then $a \mid 3b^3 - 4b^2 + 18b + 2a$.

4. Let m and n be integers. If m and n have different parity, then their sum is odd.

5. Let x be a real number. If $x^2 + 3x < 0$, then $x < 0$.

6. (Careful!) Let a and b be integers. If $a^2(b^2 - 2b)$ is odd, then a and b must both be odd.

Proofs & Counterexamples – Answers / Solutions

1. (a) True. Take $p = 97$ (or 101, 103).
 (b) This is false, since $r^2 + 2r + 2 = (r^2 + 2r + 1) + 1 = (r + 1)^2 + 1 \geq 1$.
 (c) This is true, but how can we find it? The simplest thing would be to find a number N where N , $N + 1$, $N + 2$, and so on are divisible, in turn, by 2, 3, 4, and so on. (Notice that we skipped 1 – *every* number is divisible by 1.) Maybe instead we should find a number N so that $N + 2$ is divisible by 2, $N + 3$ is divisible by 3, and so on up to $N + 101$ is divisible by 101 (so 2 through 101 give 100 numbers). In general we want $N + k$ to be divisible by k for all k from 2 through 101. This gives us our answer: just take N to be a number that is divisible by every number from 2 through 101. The easiest number (although certainly not the smallest) with this property is $N = 101!$. So the numbers

$$101! + 2, 101! + 3, 101! + 4, \dots, 101! + 101$$

are all composite.

2. (a) This is true, since $r^2 + 2r + 1 = (r + 1)^2 \geq 0$.
 (b) This is true, and we'll prove it in a number of ways:
- Directly. If n is even, then $n = 2k$ for some positive integer k and $\frac{n(n+1)}{2} = k(2k+1)$ is an integer. If n is odd, then $n = 2k - 1$ for some positive integer k and $\frac{n(n+1)}{2} = (2k-1)k$ is an integer.
 - One fact we'll see in the next few days is that

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}.$$

Written more succinctly, this is $\sum_{k=1}^n k = \frac{n(n+1)}{2}$. Thus this number is the sum of the first n integers, and so must be an integer itself.

- Another fact we'll see in the coming days is that $\frac{n(n+1)}{2}$ can also be written as $\binom{n+1}{2}$ (read “ $n + 1$ choose 2” and called a *binomial coefficient*). This is perhaps easiest to describe as *the number of ways to pick a set of 2 objects from a collection of $n + 1$ different objects*. This “number of ways” is clearly an integer.
- (c) Well, no: $p = 2$ is prime and even.

3. We provide a direct proof. If $a \mid b$, then $b = ka$ for some integer k . Then

$$3b^3 - 4b^2 + 18b + 2a = 3k^3a^3 - 4k^2a^2 + 18ka + 2a = (3k^3a^2 - 4k^2a + 18k + 2)a.$$

That is, $a \mid (3b^3 - 4b^2 + 18b + 2a)$.

4. If m is even and n is odd, then we can write $m = 2k$ and $n = 2\ell + 1$ for integers k and ℓ . Thus $m + n = 2(k + \ell) + 1$ is odd.

What if m were odd and n even? Then the argument would be exactly the same, except with the roles of m and n reversed. Usually we don't repeat the same argument, but instead

note that it doesn't really matter (for the argument) which one is even and which is odd. We typically begin our proof with something like "Without loss of generality, assume m is even and n is odd." This means that we could have assumed the opposite parities, but it doesn't really matter and the proof is the same.

5. We instead prove the contrapositive, which is equivalent: If $x \geq 0$, then $x^2 + 3x \geq 0$. This statement is nearly obvious: both x^2 and $3x$ are non-negative, so their sum is as well: $x^2 + 3x \geq 0$.
6. We'll prove the contrapositive: *if at least one of a or b is even, then $a^2(b^2 - 2b)$ is even*. There are two cases: If a is even, then $a = 2k$ and $a^2(b^2 - 2b) = 2[2k^2(b^2 - 2b)]$ is even. If b is even, then $b = 2m$ and $a^2(b^2 - 2b) = a^2(4m^2 - 4m) = 2[a^2(2m^2 - 2m)]$ is even. Of course, it could be the case that both a and b are even, so we could use either argument.